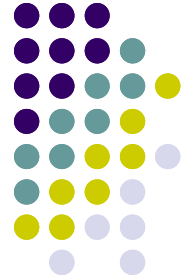


Strategies



Architecture of Real-Time systems
SE 456

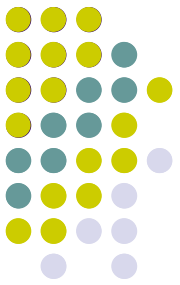
Ed Keenan

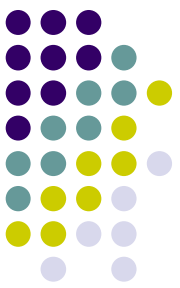
10 March 2021

13.0.8.6.1 Mayan Long Count

Goals

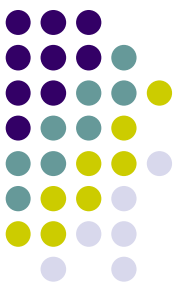
- End to End
- Evaluate
- Tools
- Wisdom





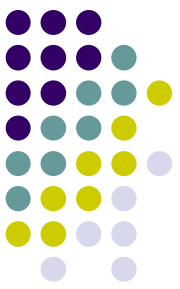
End to End

- Get your game working
 - End to End
 - Stub out major flow
 - Worry about transitions
- Get the game cycling
 - Splash
 - Start → Game → End
 - Back to start



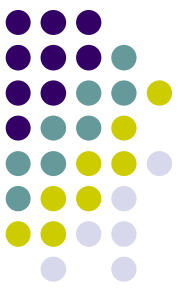
Evaluate Game

- What is the biggest problem?
 - If you had to turn it in tomorrow
 - What is the biggest missing item/feature?
 - Work on that next
- 80% done is better than 100% not done
 - Don't focus on hitting a home run
 - Let iterations be your friend
 - Stub it out → refine it



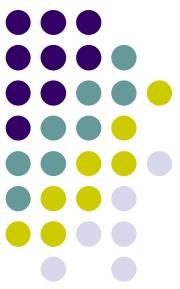
Tools Help

- Working on a dangerous system?
 - Prototype
 - Create a new project
 - Experiment
 - Explore it
 - Branch Code
 - Safe sandbox
 - Move faster/looser with modifications
 - Migrate back with version control



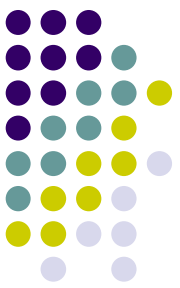
Add feature

- Added a new feature
 - Seems to work alone
 - Everything in the system is now
 - Broken or needs to be reworked
- Made one more addition
 - It's doing something different
 - Can't go back to what worked



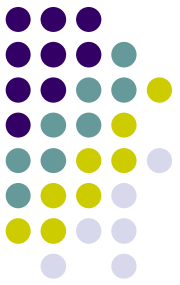
Shit! – add feature (everything breaks)

- Why?
 - Coupling
 - Orthogonality
 - Interdependency
 - Narrow vision
 - Ineffective backups
 - Running quick and loose



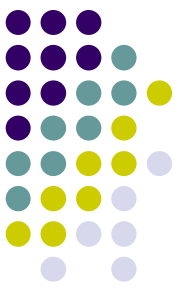
Unknowns (breaks everything)

- Paralysis of analysis
 - Thinking too hard on the problem
 - Trying to make everything work on paper first
- Hesitating to implement
 - Or Implementing too much too fast
- Better to get a 1st pass developed
 - Refactor it to a correct solution
 - Wrestling with the *Unknown*



Baby Steps or Everything breaks

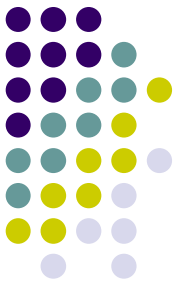
- Can't say this enough
 - *ITERATE, Iterate, iterate*
 - *Baby Steps!*
 - what about bob



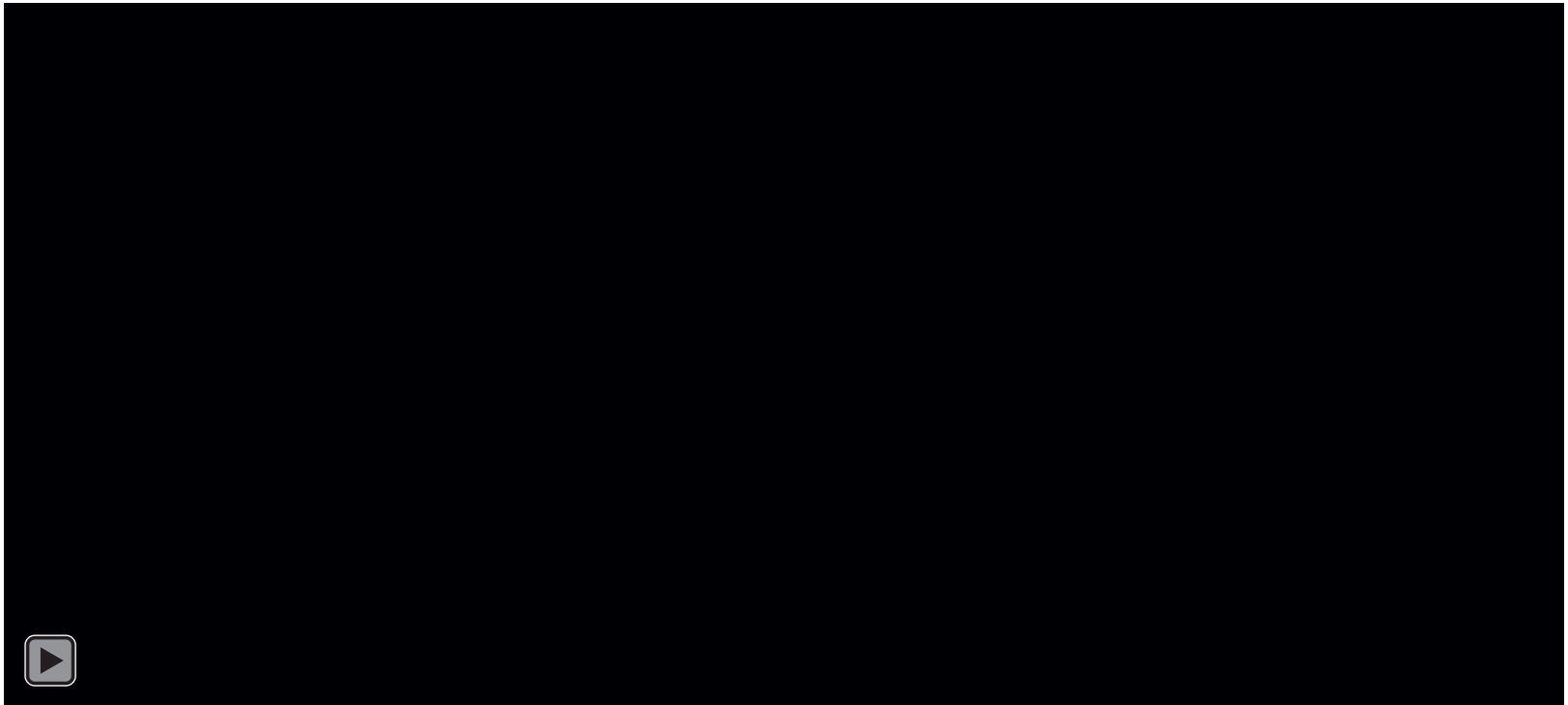
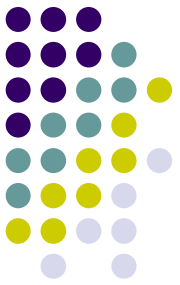
Can't Debug

- Why?
 - You need a way to see what is going on
 - You need visual clues
 - Spend time on debug features
 - Code is correct
 - You just don't understand what you are asking
- Iteration
 - Walk every line once!

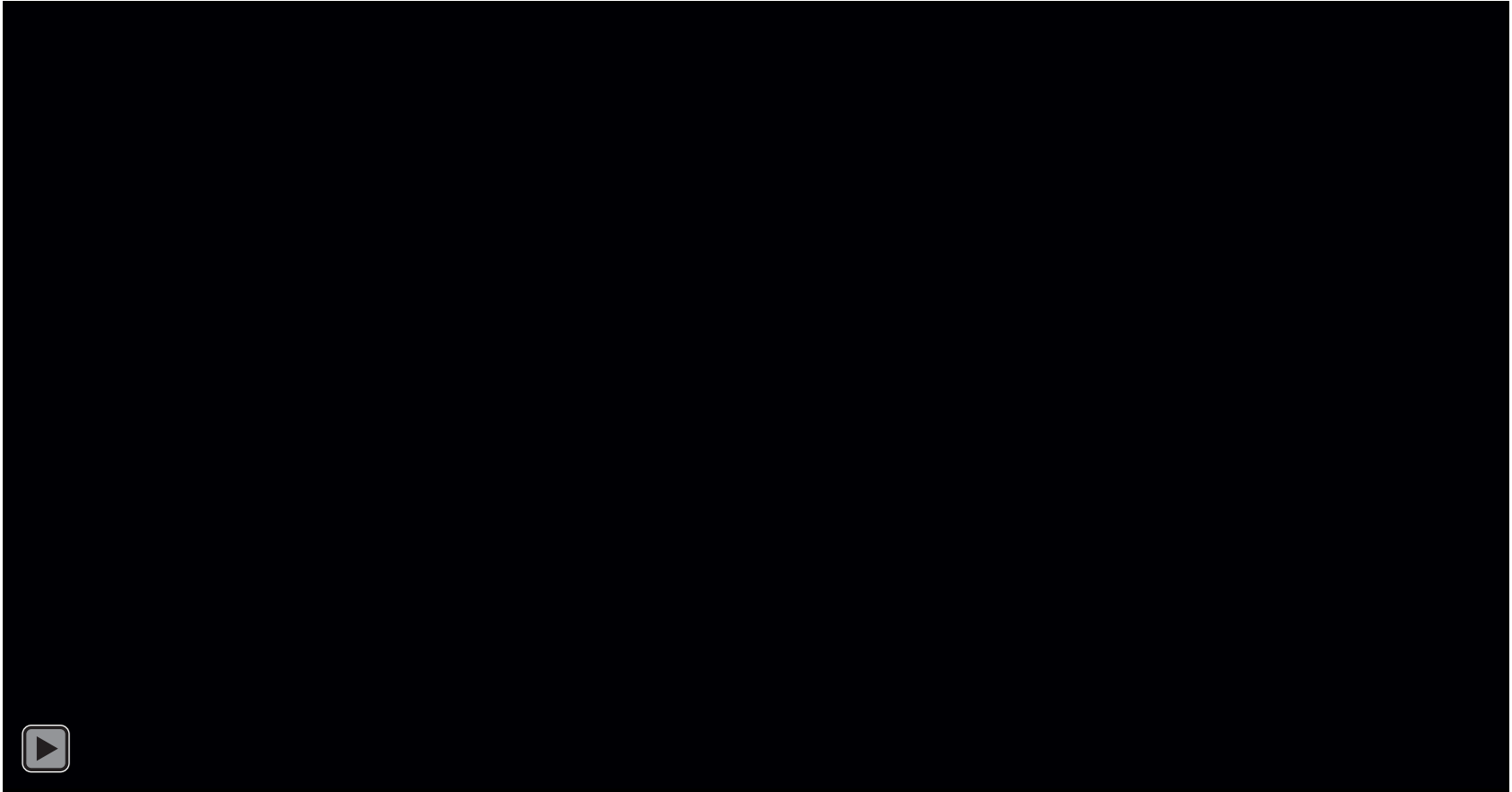
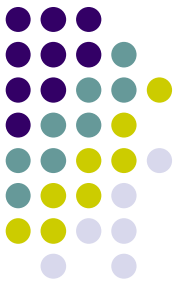
Wisdom - Fear



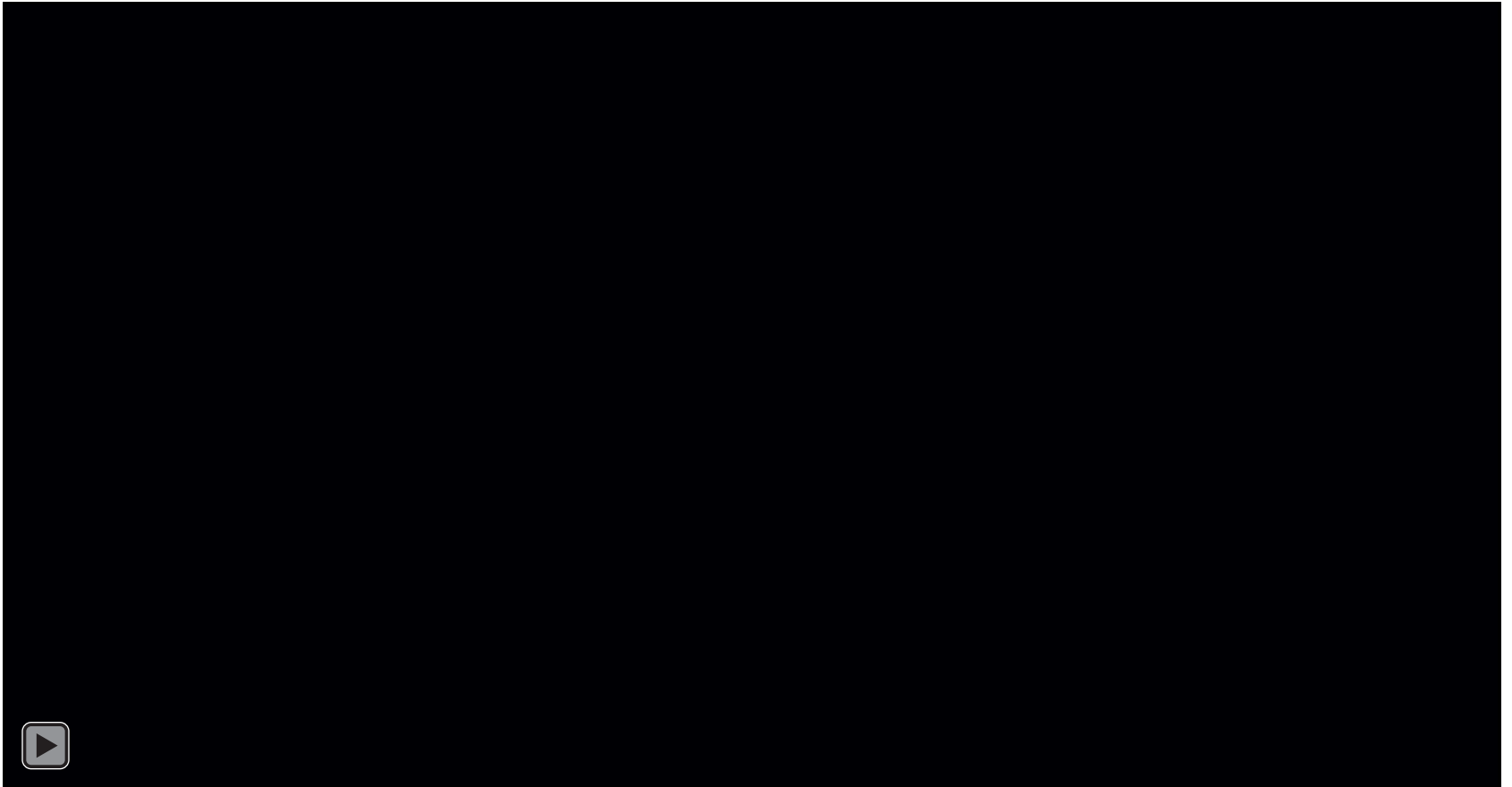
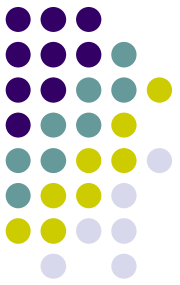
Wisdom - Unlearn

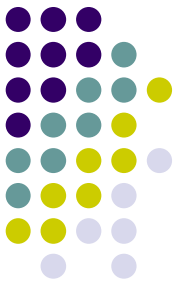


Wisdom - Control

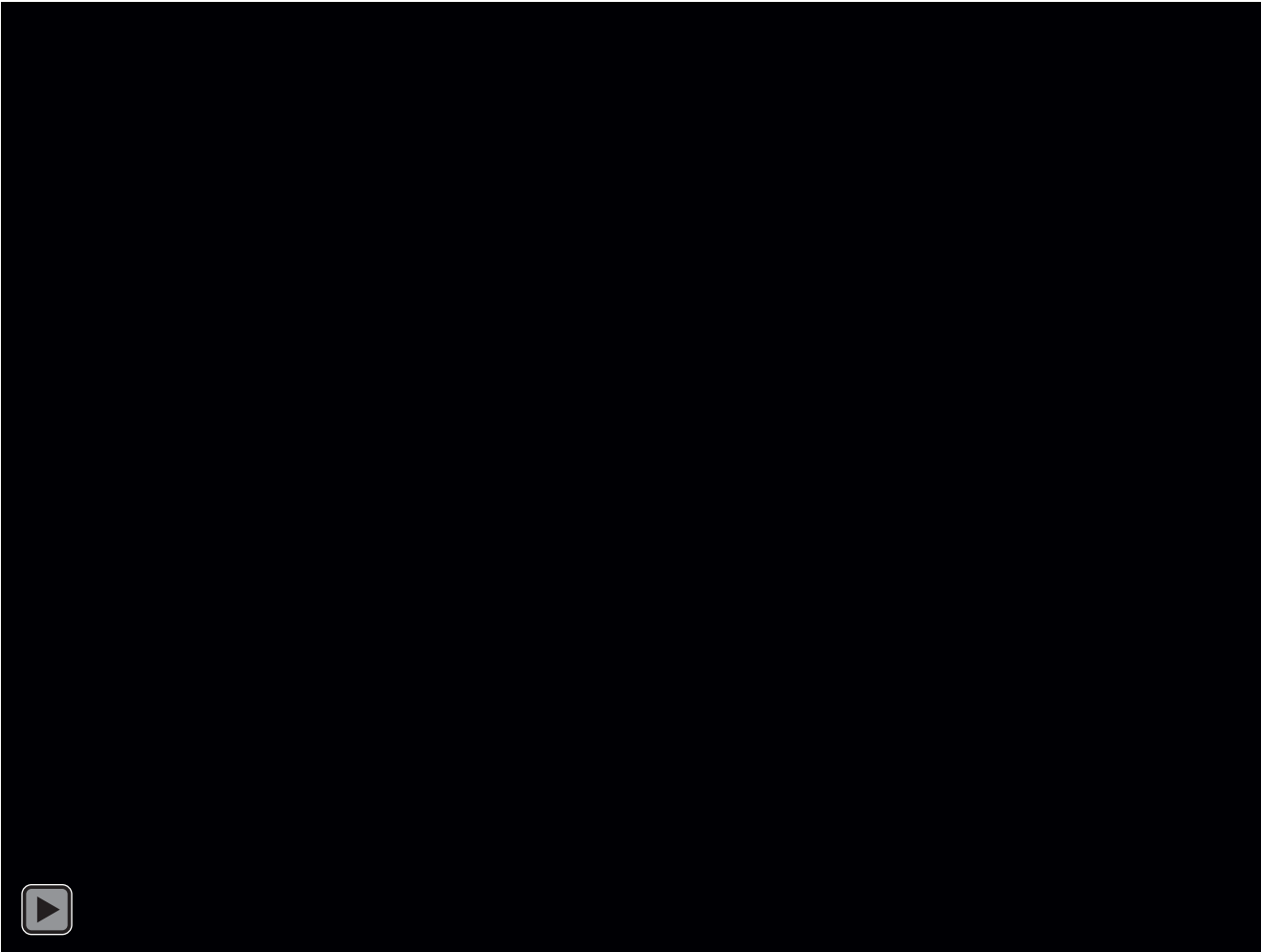


Wisdom - Do

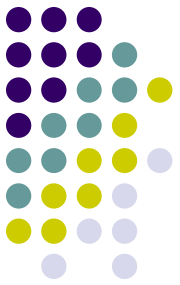




Give me help (Feature List)



Thank You!



- Questions?

